Proficiency test - 1 (20-points)

Instructions

As a reminder, this is an individual assessment. You are allowed the following resources to complete the tasks below:

– Notesheet (front + back)

– Help files in RStudio

You are **not** allowed external resources, or a LLM. This is an assessment of your understanding.

Before the lab ends, please turn this in on Gradescope.

You may not work on the problems below after your 50-minutes is up. Turn in what you have, you are able to earn partial credit on proficiency tests.

Your document should include text, all code used, and results generated from code you have created.

You may ask questions. If your question gives you an advantage over other students, it will not be answered. If you receive an error that has been covered in-class, it is your responsibility to problem solve and fix the error. If you are running into a technical error beyond the scope of the course, a TA may assist you.

I suggest rendering your document after each task.

Good luck!

Task-1

Create a new Quarto-document and include the following in the YAML:

– A title that says "Proficiency Test 1"

– A subtitle that says "ST-115"

– An author mapping with your name

You may use the following example code from the slides to help answer this question:

```
title: "Test Document"
author:
  - name: Elijah Meyer
    affiliation: NC State University
```

Task 2

Delete all the text and examples that Quarto gives you by default when making a new document. Next...

– Make a level-2 header that says Iris Dataset

– Pull up the help file for the `iris` dataset, by typing the code into your console, and copy the description of the `iris` dataset underneath your level-2 header

Task 3

Make a level-2 header that says Flower.

There are 3 species of flower.

– Compare the mean Petal Length of each of the three species of flowers (setosa versicolor virginica). Report the mean of each of the three species' Petal Length, and write a sentence saying which one is the largest.

Hint: The iris dataset is broken down in groups rows of 50 (sequential) for each flower. The first 50 are setosa. The second 50 are versicolor. The third 50 and virginica.

– Next, create a list, named `flower`, for the setosa species that has the following variable names: species, measurements. For the species variable, input the value setosa. For the measurements variable, input the mean and median value of Petal Length. Include the name `flower` in your code chunk so the results print in your rendered document.

Render your document for the final time, and turn in a PDF on Gradescope.